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Ozkan Ozturk

Academic Appointments

Giresun University, Turkey

May 2025 - present **Professor and Chair, Division of Applied Mathematics**, Department of Mathematics

Aug 2020 - May 2025 **Associate Professor**, Department of Mathematics

Dec 2017 - Aug 2020 **Research Associate**, Department of Mathematics

Appointments During Approved Leave from Giresun University

Jan 2021 - Jul 2023 **Postdoctoral Teaching Scholar**, Texas Tech University, Lubbock, TX, USA

Sep 2018 - Dec 2020 **Assistant Professor**, American University of the Middle East, Egaila, Kuwait

Earlier Appointments

Aug 2016 - May 2017 **Postdoctoral Scholar**, Missouri University of Science and Technology, Rolla, MO, USA

Aug 2012 - May 2016 **Graduate Teaching Assistant**, Missouri University of Science and Technology, Rolla, MO, USA

Education

May 2016 **Ph.D. in Mathematics**, *Missouri University of Science and Technology*, Rolla, MO, USA
Advisor: Prof. Elvan Akin

May 2016 **M.S in Electrical Engineering**, *Missouri University of Science and Technology*, Rolla, MO, USA
Advisor: Prof. Jagannathan Sarangapani

May 2011 **M.S in Applied Mathematics**, *Missouri University of Science and Technology*, Rolla, MO, USA
Advisor: Prof. Elvan Akin

Jun 2007 **B.S in Mathematics**, *Kafkas University*, Kars, Turkey

Research Interests

Nonlinear Dynamical Systems and Control: Stability analysis and adaptive, optimal, nonlinear, and machine-learning-assisted control of continuous-time, discrete-time, sampled-data, and hybrid systems, with emphasis on unified time-scale formulations and applications to unmanned aerial vehicles (UAVs), unmanned ground vehicles (UGVs), and other mobile robotic platforms.

Mathematical and Computational Biology: Dynamical modeling, stability and sensitivity analysis, parameter estimation, data fitting, and control of infectious-disease and physiological systems, with applications to tuberculosis, HIV infection, and heart-rate regulation.

Qualitative Theory of Dynamic Equations: Oscillation, nonoscillation, stability, and asymptotic analysis of higher-order, delay, and Volterra integro-dynamic equations on time scales.

Publications and Manuscripts

Manuscripts Under Review

[M2] E. Akin, **Ö. Öztürk** and N. Pelen. A General-Incidence SEIR Model Developed for Tuberculosis with Latent Recovery: Stability, Sensitivity, and Data Fitting, under review

[M1] **Ö. Öztürk**, T. Cuchta, and N. Wintz. Lyapunov stability conditions on a conformable heart rate model, under review.

Peer-Reviewed Journal Articles

[J20] **Ö. Öztürk** and A. Akgül. Control of Unmanned Ground Vehicles with Fractional Derivative, Journal of Vibration Testing and System Dynamics, forthcoming, January 2027.

[J19] **Ö. Öztürk**, E. Akin, and N. Pelen. Nonoscillatory Solutions of Third-Order Nonlinear Dynamic Equations: Existence and Nonexistence, Turkish Journal of Mathematics, 49(5), 2025.

[J18] **Ö. Öztürk** and R. Higgins. Almost Oscillation of a Third-Order Emden-Fowler Equation on Time Scales, Differ. Equ. Dyn. Syst., Volume 32, 979–999, 2022.

[J17] **Ö. Öztürk**, R. Higgins and G. Kittou. Oscillation of Three Dimensional Time Scale Systems with Fixed Point Theorems, Filomat, 36:5, 2021.

[J16] **Ö. Öztürk** and H. M. Güzey. Saturation Control of PVTOL Aircraft on Time Scales, Journal of Mathematics and Computer Science, 21:3, 2020.

[J15] E. Akin, H. M. Güzey and **Ö. Öztürk**. Control of Wheeled Mobile Robots on Time Scales, Differ. Equ. Dyn. Syst, 31, 199–208, 2019.

[J14] E. Akin, T.Hassan, **Ö. Öztürk** and I.U. Tiryaki. On nonoscillatory solutions of three dimensional time-scale systems, Turk J. Math., 43: 2246 – 2261, 2019.

[J13] E. Akin and **Ö. Öztürk**. On Volterra-Integro Dynamical Systems on Time Scales, Communications in Applied Analysis, Volume 23, No. 1, 21-30, 2019.

[J12] **Ö. Öztürk** and H. M. Guzey. Optimal Control of Quadrotor Unmanned Aerial Vehicles on Time Scales, Int. J. Difference Equ., 1:13, 2018.

[J11] **Ö. Öztürk** and R. Higgins. Limit behaviors of Nonoscillatory Solutions of Three-Dimensional Time Scale Systems, Turk J. Math., 42, 2576-2587, 2018.

[J10] **Ö. Öztürk**. On Oscillatory Behavior of Two-Dimensional Time-Scale System, Advances in Difference Equations, 2018:18, 2018.

[J9] **Ö. Öztürk**. On Oscillation of Two-Dimensional Time Scale Systems with a Forcing Term, Turk J. Math., 42: 312-319, 2018.

[J8] E. Akin, **Ö. Öztürk**, I. U. Tiryaki, and G. Yeni. A Note on Nonoscillatory Solutions for Higher Dimensional Time Scale Systems, Int. J. Appl. Exp. Math., 3: 126, 2018.

[J7] **Ö. Öztürk**. On the Existence of Nonoscillatory Solutions of Three-Dimensional Time Scale Systems, J. Fixed Point Theory Appl. 19 2617-2628, 2017.

[J6] **Ö. Öztürk**. Classification Schemes of Nonoscillatory Solutions for Two-Dimensional Time Scale Systems, Math. Inequal. Appl., Volume 20, Number 2, 377-387, 2017.

[J5] E. Akin and **Ö. Öztürk**. Limiting Behaviors of Nonoscillatory Solutions for Two-Dimensional Nonlinear Time Scale Systems, Mediterr. J. Math., 14:34, 2017.

[J4] **Ö. Öztürk**, E. Akin, and I. U. Tiryaki. On Nonoscillatory Solutions of Emden-Fowler Dynamic Systems on Time Scales, Filomat 31:6, 1529-1541, 2017.

[J3] **Ö. Öztürk** and E. Akin. On Nonoscillatory Solutions of Two-Dimensional Nonlinear Delay Dynamical Systems, Opuscula Mathematica, vol. 36, no. 5, 2016.

[J2] **Ö. Öztürk** and E. Akin. Nonoscillation Criteria for Two-Dimensional Time-Scale Systems, Nonauton. Dyn. Syst. 3:1-13, 2016.

[J1] **Ö. Öztürk** and E. Akin. Classification of Nonoscillatory Solutions of Emden-Fowler Dynamic Equations on Time Scales, Dynamic Systems and Applications, 25, 219-236, 2016.

Refereed Book Chapters

[B3] **Ö. Öztürk**. Existence and Asymptotic Behaviors of Nonoscillatory Solutions of Third Order Time Scale Systems. Advances in the Solution of Nonlinear Differential Equations, ISBN 978-1-83968-657-3, 2021.

[B2] **Ö. Öztürk**. Oscillation Criteria of Two-Dimensional Time-Scale Systems. Oscillators-Recent Developments, ISBN 978-1-78985-838-9, 2019.

[B1] E. Akin and **Ö. Öztürk**. On Nonoscillatory Solutions of Two-Dimensional Nonlinear Dynamical Systems. Dynamical Systems-Analytical and Computational Techniques, ISBN 978-953-51-3015-4, 2017.

Research Projects, Funding, and International Collaboration

Funded Research Project

2018 - 2021 **ERA-NET/TÜBİTAK Project No. 217E138**: *Collaborative Strategies for Heterogeneous Robots in Agricultural Missions Using Intuitive Human–Robot Interfaces*. Role: Researcher. Total project funding: ₺367,000 (approximately US$69,904 at the time of the award).

International Research Networks (European Union-Funded COST Actions)

2025 - 2029 **COST Action CA24136:** *Interactions Between Control Theory and Machine Learning*. Role: Working Group Member (WGs 1, 2, and 5).

2025 - 2029 **COST Action CA24104:** *Stochastic Differential Equations: Computation, Inference, Applications*. Role: Working Group Member (WGs 1, 2, 3, and 5).

2025 - 2029 **COST Action CA24133:** *European Network for Animal Evolutionary Immunology–Interlinking Laboratory and Wildlife Infection Biology*. Role: Working Group Member (WGs 1, 3, 4, and 5).

2025 - 2029 **COST Action CA24140:** *One Health Zoonotic Hepevirus Network*. Role: Working Group Member (WGs 1–4).

Research Mobility, Travel Grants, and Scholarships

Sep 2026 **Research Mobility (Short-Term Scientific Mission):** "Dynamical Equations on Time Scales as a Control-Theoretic Framework for Analyzing Adaptive Learning in Autonomous Robotic Systems", Budapest, Hungary; supported by COST Action CA24136: €1,700

Jul 2026 **Travel Grant:** 15th AIMS Conference on Dynamical Systems, Differential Equations and Applications, Athens, Greece; supported by COST Action CA24136 (€1,450).

Apr 2026 **Travel Grant:** InterCoML Conference NextGen Synergy; supported by Cost Action CA24136: €991.51

2018, 2024 **Travel Grant:** TÜBİTAK: $2,000 (2018) and $1,500 (2024).

2014 - 2016 **Travel Grant:** SEARCDE: $1,000 each in 2014, 2015, and 2016.

2007 - 2016 **Graduate Scholarship:** awarded by the Ministry of National Education of Turkey for master's and doctoral study in the United States. Total support: approximately $250,000.

Conference Organizations and Presentations

July 2026 **The 15th American Institute of Mathematical Sciences**, Athens, Greece

Special Session Organizer, Advances in Differential, Difference and Dynamic Equations with Applications in Science and Engineering

Presentation: *Existence Results for Third-Order Nonlinear Dynamic Equations*.

Presentation: *Conformable Control of Unmanned Ground Vehicles*, in the contributed session *ODEs and Applications*.

July 2026 **The 31st International Conference on Difference Equations and Applications**, Milan, Italy

Presentation: *Understanding Tuberculosis Dynamics through a General SEIR Model*, in the special session *Recent Advances in Time Scales Calculus*.

Apr 2026 **InterCoML Conference: NextGen Synergy**, Prague, Czech Republic

Presentation: *Control of Dynamical Systems on Time Scales: Bridging Continuous and Discrete Learning Frameworks*.

Jun 2024 **The 29th International Conference on Difference Equations and Applications**, Paris, France

Presentation: *Classification of Nonoscillatory Solutions of a Third Order Nonlinear Dynamic Equation*, in the special session *Recent Advances on Time Scales and Its Relation to Difference Equations*.

Apr 2022 **Joint Mathematics Meetings**, Seattle, WA, USA

Special Session Organizer, Analysis of and Recent Advances in Difference, Differential and Dynamic Equations with Applications

Jul 2018 **The 12th AIMS Conference on Dynamical Systems, Differential Equations and Applications**, National Taiwan University, Taipei City, Taiwan

Presentation: *Recent Advances on Nonoscillatory Solutions of Two-Dimensional Time-Scale Systems*, in the special session *Advances in Qualitative Theory of Differential, Difference, and Dynamic Equations*.

Presentation: *Control of Mobile Robots on Time Scales*, in the contributed session *Control and Optimization*.

Jun 2018 **ECCO 2018**, University of Fribourg, Fribourg, Switzerland

Presentation: *Optimal Control of Unmanned Aerial Vehicles on Time Scale Theory*.

Nov 2017 **Costas 2017**, Samsun, Turkey

Presentation: *Control of Unmanned Ground Vehicles on Time Scales*.

Jan 2017 **Joint Mathematics Meetings**, Atlanta, GA, USA

Presentation: *Optimal Control of Unmanned Aerial Vehicles*.

Nov 2016 **The 36th SEARCDE**, Fort Myers, FL, USA

Presentation: *Classification Schemes for Nonoscillatory Solutions of Nonlinear Dynamical Systems on Time Scales*.

Aug 2016 **MAA Math Fest**, Columbus, OH, USA

Presentation: *On Nonoscillatory Solutions of Two Dimensional Nonlinear Delay Dynamical Systems*.

Oct 2015 **Fall Southeastern Sectional Meeting**, - University of Memphis, Memphis, TN, USA

Presentation: *Classification of Nonoscillatory Solutions of Dynamical Systems on Time Scales*.

Oct 2015 **The 35th SEARCDE**, Greensboro, NC, USA

Presentation: *On Nonoscillatory Solutions of Systems of First Order Dynamic Equations on Time Scales*.

May 2015 **The 7th International Conference on Dynamic Systems and Applications and 5th International Conference on Neural, Parallel, and Scientific Computations**, Atlanta, GA, USA

Presentation: *On Nonoscillatory Solutions of Emden Fowler Dynamic Systems on Time Scales*.

Apr 2015 **The 1st Meeting of SIAM Central States Section**, Missouri S&T, Rolla, MO, USA

Presentation: *On Nonoscillatory Solutions of Emden Fowler Dynamic Systems on Time Scales*.

Mar 2015 **MAA Missouri Section Spring Meeting**, Rolla, MO, USA

Presentation: *Classification of Nonoscillatory Solutions of Emden-Fowler Dynamic Equations on Time Scales*.

Oct 2014 **The 34th SEARCDE**, University of Memphis, Memphis, TN, USA

Presentation: *Asymptotic Properties of Solutions of Emden- Fowler Dynamic Equations*.

Apr 2014 **AMS 1100 Sectional Meeting**, Texas Tech University, Lubbock, TX, USA

Presentation: *Asymptotic Behavior of Emden–Fowler Dynamic Equations on Time Scales*, in the special session *Analysis and Applications of Dynamic Equations on Time Scales*.

Graduate Student Supervision and Committees

M.S. Thesis Advisor

2025 - present **Yekbun Akman**, "Mathematical Modeling and Control of Unicycle-Type Mobile Vehicles"

2024 - 2026 **Abdulkadir Yilmaz**, "Mathematical Modeling and Stability Analysis of the Monkeypox Virus" *(defended in March 2026)*

M.S. Thesis Committee Member

Jun 2025 **Ilayda Altinok**, "Biocontrol of Brown Marmorated Stink Bugs Using Samurai Wasps: Investigating the Impact on Hazelnut Production in Türkiye"

Professional Service and Leadership

2018 - 2019 **Editor**, *Expert Opinion on Astronomy and Astrophysics*; and *International Journal of Mathematics and Systems Science*.

2018 - present **Peer Review**, Mathematical Reviews/MathSciNet; Journal of Mathematical Biology; Differential Equations and Dynamical Systems; Communications Faculty of Sciences University of Ankara Series A1; Journal of Inequalities and Special Functions; Journal of Inequalities and Applications; Journal of Applied Mathematics and Computing; and Mathematics and Statistics.

2012 - 2017 **Leadership**, Turkish Student Association, Missouri University of Science and Technology: Advisor (2016–2017), President (2014–2015), and Vice President (2012–2014).

Teaching Experience

2023 - present **Giresun University**, Instructor of Record

Graduate courses: Nonlinear Differential Equations I–II; Selected Topics in Differential Equations I–II; Advanced Partial Differential Equations I; LaTeX.

Undergraduate courses: Differential Equations I–II; Differential Equations for Engineers; Numerical Analysis I–II; Partial Differential Equations I; Complex Analysis; Calculus II for Engineers; Advanced Programming I–II (Python).

2021 - 2023 **Texas Tech University**, Instructor of Record

Courses taught: Linear Algebra; Calculus II–III with Applications; Higher Mathematics for Engineers and Scientists I.

2018 - 2020 **American University of the Middle East**, Instructor of Record

Courses taught: Calculus and Analytic Geometry I–III; Linear Algebra; Ordinary Differential Equations; Ordinary and Partial Differential Equations.

2016 - 2017 **Missouri University of Science and Technology**, Lead Lecturer

Courses taught: Calculus for Engineers I; College Algebra.

2012 - 2016 **Missouri University of Science and Technology**, Graduate Teaching Assistant

Courses taught: Elementary Differential Equations; Calculus for Engineers II; Trigonometry.

Honors and Awards

Research Honors

Apr 2021 Distinguished Academic Achievement, Texas Tech University, USA

Jun 2018 Faculty Excellence Award in Research, Giresun University, Türkiye

Jun 2016 Institutional Nominee, 2016 CGS/ProQuest Distinguished Dissertation Award, Missouri S&T, USA

Teaching Honors

May 2017 We Love Your Class Award, 11th Annual Freshman Engineering Program, Missouri S&T, USA

Sep 2014 Most Honorable Mention Award, Department of Mathematics and Statistics, Missouri S&T, USA

Dec 2013 Teaching Excellence Award, Department of Mathematics and Statistics, Missouri S&T, USA

Service and Leadership Honors

Jan 2015 Student Life Honor List for Fall 2014, Missouri S&T, USA

Oct 2013 Best Officer Award, Turkish Student Association, Missouri S&T, USA

Technical Skills, Certificates and Languages

Python, C++, MATLAB LaTeX

May 2019 MATLAB Certificate by Vanderbilt University, USA

Mar 2018 Control of Nonlinear Spacecraft Attitude Motion Certificate by University of Colorado, USA

Sep 2016 Control of Mobile Robots Certificate by Georgia Institute of Technology, USA

May 2009 Intensive English Language Program Certificate of Completion by Saint Joseph University, USA

Languages Turkish(native), English(advanced), German and French(beginner)